NCCP 00455 · VERSION 16

Your treatment, explained

First-line metastatic MSI-H / dMMR colorectal cancer · NCCP 00455 v16 · Document revised 5 November 2026

Your personal plan: this is the standard regimen schedule. Your team confirms your prescribed doses, cycles, treatment changes and appointments in your written plan.

Pembrolizumab 200 mg · colorectal

21-day cycle | Source reconciled; independent clinical, pharmacy and patient/PPI review pending

ACT ON SYMPTOMS, NOT THE CALENDAR

When to get help

Emergency: call 112 / 999

Severe breathing difficulty, collapse, seizure or severe chest pain: call emergency services. Do not wait for an online answer.

Contact your oncology service immediately

Temperature 37.5°C or higher, 36°C or lower, shivering or suddenly feeling unwell — even with a normal temperature. Contact oncology immediately for new breathlessness, persistent diarrhoea, severe tummy pain, yellow skin or eyes, much less urine, spreading or blistering rash, severe headache, marked weakness, confusion, unusual thirst or passing much more urine. Severe breathing difficulty, chest pain, collapse or seizure needs emergency care. Effects can begin after treatment ends.

Follow the urgent contact instructions from your treating hospital. If you cannot reach the team, seek urgent medical assessment. Outside clinic hours, use your instructed out-of-hours pathway or the emergency department.

This page does not diagnose symptoms or decide whether it is safe to stay at home. You do not need to know which medicine caused a symptom before calling.

Your treatment and urgent-contact card is supplied separately and in the wallet-card section of the complete handbook.

Sources: hse · nccp · smpc

Reference contents

YOUR COMPLETE REFERENCE

Read what you need, when you need it.

Start with the overview, explore the cycle, or go straight to a question. Urgent advice is always available.

How each drug works
Treatment stages
Your 21-day timeline
Baseline and cycle-specific checks
Whole treatment, scans and checks
Tablets, visits and practical planning
Your colorectal cancer and treatment aim
How pembrolizumab works
Your treatment course
Tests and follow-up
Eating, energy and daily life
When to get urgent help
Supportive medicines
Understanding the evidence
Questions to ask
Sources

UNDERSTAND YOUR TREATMENT

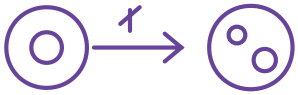
Three medicines. Different jobs.

Pembrolizumab aims to control bowel cancer that has spread and has a particular DNA-repair feature, called MSI-H or dMMR. Your team discusses likely benefit, uncertainty and alternatives for you.

IMMUNOTHERAPY

Pembrolizumab

Keytruda



Blocks an immune brake

Pembrolizumab is an antibody that blocks PD-1, a control switch on immune T cells. Cancer cells can use this switch to reduce immune attack. Blocking it can help the immune system recognise and attack cancer. The same increased immune activity can inflame healthy organs, which is why new symptoms need attention even after treatment ends.

Immune cell → PD-1 blocked → Cancer attacked

What to watch for

Report new breathing, bowel, skin, hormone, heart or nerve symptoms promptly; these may be immune inflammation.

Sources: smpc

The drawings are simplified explanations of drug action, not anatomical pictures or a prediction of response.

The anatomical side-effect map follows.

REGIMEN-SPECIFIC SIDE EFFECTS

Your anatomical side-effect map

A picture of side effects

Match each symptom box to its numbered body location. Tell your team about new symptoms.

1 Hormones / nervous system

Pembrolizumab

Severe headache, confusion, weakness, dizziness or marked thirst need prompt advice.

3 Heart

Pembrolizumab

Chest pain, racing heartbeat or fainting needs urgent assessment.

5 Kidneys

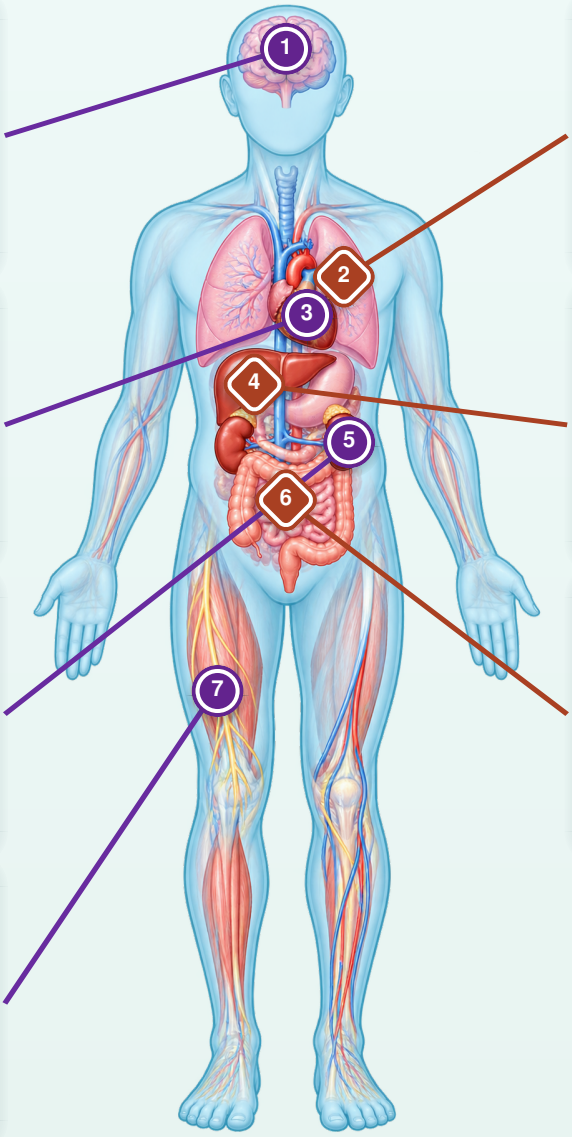
Pembrolizumab

Report much less urine, new swelling or a change in kidney tests.

7 Skin / muscles

Pembrolizumab

Spreading rash, blisters, severe weakness or muscle pain needs prompt review.



2 Lungs

Pembrolizumab

New cough or breathlessness can signal inflammation or another urgent cause.

4 Liver

Pembrolizumab

Yellow eyes/skin or dark urine may signal inflammation; tests also check the liver.

6 Bowel

Pembrolizumab

New or persistent diarrhoea, blood in stools or severe tummy pain needs urgent advice.

Selected effects, not every possible effect or a prediction. Blood-count, skin and line markers are schematic. Number and text carry meaning; colour is decorative.

THE BIGGER TREATMENT PLAN

Treatment → review → next steps

ONE IMMUNOTHERAPY MEDICINE

Pembrolizumab every 3 weeks

C1 C2 C3 ...

Continue while treatment helps and side effects remain acceptable. At 24 months, the team may consider stopping if there is a sustained response without progression; this is an individual review, not an automatic fixed final cycle.

Your team confirms each stage and discusses any changes with you.

Sources: nccp

BASELINE → TREATMENT → FOLLOW-UP

Checks across your colorectal treatment

There is no fixed NCCP scan or tumour-marker interval in this regimen. The team individualises imaging, markers and extra tests.

The team arranges and reviews tests. This guide shows expected timing; it does not record completed appointments. NCCP 00455 v16 official regimen

THE WHOLE TREATMENT JOURNEY

Pembrolizumab 200 mg · colorectal: treatment and checks

First-line metastatic MSI-H / dMMR colorectal cancer

Pembrolizumab 200 mg is given over about 30 minutes on Day 1 every 3 weeks (21 days). The visit takes longer for assessment and preparation.

Read from left to right. Columns show stages, not equal lengths of time. This is a general plan: no dates or personal information are needed. Your team confirms which steps apply and arranges appointments.

Treatment, tests and reviews across the course

Care / check	Before starting	Infusion visits	Between visits	Course review	After treatment
Blood count, kidneys, liver and glucose NCCP 00455	Baseline blood tests	Before every treatment cycle	Extra tests for new symptoms or abnormalities	Clinical review before restart after a hold	Follow abnormal results to resolution

Care / check	Before starting	Infusion visits	Between visits	Course review	After treatment
Thyroid and other hormones NCCP 00455	Baseline thyroid function	Check thyroid timetable	Thyroid every 3–6 weeks; cortisol if indicated	Confirm any hormone replacement plan	Hormone treatment may continue long term
Symptoms and immune inflammation NCCP 00455	Discuss autoimmune disease, infection, medicines and previous immunotherapy	Review new symptoms before infusion	Report symptoms immediately; do not wait for the next cycle	Pauses or stopping are clinician decisions	Immune effects can occur after the last dose
Cancer response and duration NCCP 00455	Confirm validated MSI-H/dMMR result and first-line setting	Agree the response assessment plan	Scans and markers as directed by the consultant	Individual review, including possible stop after sustained response at 24 months	Agree follow-up and the contact pathway

What each check is for

1

Pembrolizumab every 3 weeks

One immunotherapy medicine

Pembrolizumab aims to control bowel cancer that has spread and has a particular DNA-repair feature, called MSI-H or dMMR. Your team discusses likely benefit, uncertainty and alternatives for you.

Ask / act: Confirm the current phase, prescription, tests and support medicines.

Sources: NCCP 00455

At the next review, ask

- Which treatment phase am I in, and what happens next?
- Which test is expected next, who arranges it, and who explains the result?
- Who should I contact if an expected appointment has not arrived?

This map does not show whether a test has been ordered, booked or completed. Do not wait for a scheduled test if you develop new symptoms.

Sources checked 2026-11-05 · Source reconciled; independent clinical, pharmacy and patient/PPI review pending · SACTCheck v0.84.35

YOUR CYCLE, DAY BY DAY

Your 21-day cycle.

Read the timing and symptom notes below. These are possible patterns, not a promise about when symptoms start or stop.

Treatment timing: displayed days 1–21

Treatment or check	Days in this window	What happens
Infusion	Day 1	Day 1 · 200 mg infusion
Immune effects	Days 1–21	Any day, and after the final dose
Symptoms and thyroid checks	Days 1–21	Thyroid tests every 3–6 weeks; report symptoms immediately

The cycle shows appointment timing. It does not show a predictable immune-toxicity peak.

The next cycle starts 21 days after this one, if your team confirms it.

> Immune effects can happen on any day and after treatment ends. Carry your immunotherapy alert card and tell any healthcare team that you receive pembrolizumab.

Complete timing and symptom notes

Day 1: Infusion and assessment

The team reviews blood results, symptoms and medicines. Tell the nurse immediately about itching, flushing, tight chest, dizziness or suddenly feeling unwell. Confirm the next tests, treatment visit and contact plan.

Day 2–21: Between infusions

Immune effects may start at any time. A long interval between treatment visits is not a break from symptom reporting. Keep thyroid tests every 3–6 weeks, including separate appointments if required.

Sources: nccp

PRACTICAL TREATMENT GUIDE

Practical treatment planning

Agree the arrangements with your oncology team.

STEP 1

Appointments

Pembrolizumab 200 mg is given over about 30 minutes on Day 1 every 3 weeks (21 days). The visit takes longer for assessment and preparation.

STEP 2

Course length

Continue while treatment helps and side effects remain acceptable. At 24 months, the team may consider stopping if there is a sustained response without progression; this is an individual review, not an automatic fixed final cycle.

STEP 3

Blood and thyroid appointments

Blood count, kidney/liver function, glucose and thyroid function at baseline. Blood count, kidney/liver and glucose before each 3-weekly cycle; thyroid every 3–6 weeks regardless of infusion interval; cortisol if indicated. Scans and markers follow the consultant plan.

STEP 4

Home medicines and records

There is no take-home chemotherapy pump. Keep your alert card, symptom record and medicine list. Discuss travel, work and transport needs.

STEP 5

After the final dose

Keep reporting new symptoms. Ask who follows persistent side effects and any hormone medicines.

Sources: NCCP 00455 v16 official regimen. Source reconciled; independent clinical, pharmacy and patient/PPI review pending.

MAKE THE PLAN PRACTICAL

Support around the medicines

1 Routine medicines

Routine premedication is not usually required. Pembrolizumab has minimal emetogenic potential; the team individualises anti-sickness treatment if needed.

2 Steroids and hormone replacement

If immune effects develop, the team may prescribe steroids or hormones. Follow the exact plan; do not stop steroids abruptly or restart pembrolizumab yourself. If treatment is paused because of an immune side effect, your team checks recovery and any steroid or hormone plan before deciding whether it can restart. Do not change steroids or restart treatment yourself.

3 Medicines and pregnancy

Bring prescription, non-prescription and herbal medicines. Discuss pregnancy prevention; effective contraception is required during pembrolizumab and for at least four months after the last dose. Do not breastfeed during treatment.

Sources: nccp · smpc

TREATMENT REFERENCE · 01

Your colorectal cancer and treatment aim

The tumour test and treatment setting matter.

MSI-H and dMMR

These tests identify a problem with the cancer's DNA-repair system. They help identify colorectal cancers more likely to benefit from this immunotherapy. Ask whether genetics assessment is relevant to you.

Sources: nccp · smpc

First-line metastatic treatment

Pembrolizumab aims to control bowel cancer that has spread and has a particular DNA-repair feature, called MSI-H or dMMR. Your team discusses likely benefit, uncertainty and alternatives for you. First-line means the first systemic treatment for the metastatic setting.

Sources: nccp · smpc

Your choices

Ask about other treatments, clinical trials and symptom-focused care, how benefit will be assessed and what matters to your quality of life.

Sources: nccp · smpc

TREATMENT REFERENCE · 02

How pembrolizumab works

It releases a brake on immune cells.

PD-1 blockade

Pembrolizumab is an antibody that blocks PD-1, a control switch on immune T cells. Cancer cells can use this switch to reduce immune attack. Blocking it can help the immune system recognise and attack cancer. The same increased immune activity can inflame healthy organs, which is why new symptoms need attention even after treatment ends.

Sources: nccp · smpc

Healthy tissues can be affected

The same immune activation can inflame the bowel, lungs, liver, kidneys, skin, hormone glands, heart, nerves or muscles. Some effects require hospital care, steroids or lasting hormone replacement.

Sources: nccp · smpc

A pause is a safety decision

Doses may be delayed or treatment stopped for immune effects. The standard approach is to hold or stop treatment, not simply to give a smaller dose.

Sources: nccp · smpc

TREATMENT REFERENCE · 03

Your treatment course

Visit frequency and total course length are separate decisions.

Infusions

Pembrolizumab 200 mg is given over about 30 minutes on Day 1 every 3 weeks (21 days). The visit takes longer for assessment and preparation.

Sources: nccp · smpc

Course reviews

Continue while treatment helps and side effects remain acceptable. At 24 months, the team may consider stopping if there is a sustained response without progression; this is an individual review, not an automatic fixed final cycle.

Sources: nccp · smpc

Evidence for the schedule

KEYNOTE-177 tested 200 mg every three weeks. The 400 mg six-week schedule is supported by exposure bridging and product authorisation; it was not a separate randomised colorectal comparison.

Sources: keynote177 · pk-bridge · smpc

TREATMENT REFERENCE · 04

Tests and follow-up

Blood tests can identify a problem before symptoms appear.

Before starting

Blood count, kidney/liver function, glucose and thyroid function at baseline. Blood count, kidney/liver and glucose before each 3-weekly cycle; thyroid every 3–6 weeks regardless of infusion interval; cortisol if indicated. Scans and markers follow the consultant plan.

Sources: nccp · smpc

Scans and tumour markers

The consultant sets the disease-assessment plan. This regimen does not prescribe a universal CT or CEA interval. A scan schedule from a research study is not automatically your schedule.

Sources: nccp

During a hold and after stopping

Keep follow-up appointments and the agreed steroid or hormone plan. Continue reporting symptoms after the last dose; another clinician needs to know you have received immunotherapy.

Sources: nccp · smpc

TREATMENT REFERENCE · 05

Eating, energy and daily life

Report changes early.

Bowel changes

Report new or persistent diarrhoea, blood in stools or abdominal pain promptly. Do not treat ongoing diarrhoea as a minor stomach upset or rely on self-treatment without advice.

Sources: nccp · smpc

Energy and hormones

New marked fatigue, headache, dizziness, feeling unusually cold, unusual thirst or passing more urine may need hormone and glucose tests.

Sources: nccp · smpc

Medicines, pregnancy and feeding

Bring prescription, non-prescription and herbal medicines. Discuss pregnancy prevention; effective contraception is required during pembrolizumab and for at least four months after the last dose. Do not breastfeed during treatment.

Sources: nccp · smpc

TREATMENT REFERENCE · 06

When to get urgent help

Do not wait for the next infusion.

Warning symptoms

Contact oncology immediately for new breathlessness, persistent diarrhoea, severe tummy pain, yellow skin or eyes, much less urine, spreading or blistering rash, severe headache, marked weakness, confusion, unusual thirst or passing much more urine. Severe breathing difficulty, chest pain, collapse or seizure needs emergency care. Effects can begin after treatment ends.

Sources: nccp · smpc

During the infusion

Tell the nurse immediately about itching, flushing, breathlessness, chest tightness, dizziness or feeling suddenly unwell.

Sources: nccp · smpc

Carry the alert card

Show your treatment card to any healthcare team, even after immunotherapy ends. Keep your actual daytime and out-of-hours oncology numbers accessible.

Sources: nccp · smpc

Evidence in MSI-H/dMMR colorectal cancer

KEYNOTE-177 compared pembrolizumab 200 mg every three weeks with chemotherapy in previously untreated MSI-H/dMMR metastatic colorectal cancer.

The trial showed a longer time before cancer worsened with pembrolizumab. This is a group result, not a guarantee for one person; ask your team about benefits, risks and alternatives in your situation.

Sources: keynote177 · keynote177-final · pk-bridge

PREPARE FOR THE CONVERSATION

Questions worth asking

Is my cancer confirmed MSI-H or dMMR, and is this the first-line metastatic plan?

What benefit and uncertainty apply to me?

Who books and reviews my blood and thyroid tests?

When are scans or tumour markers planned?

Which symptoms need immediate advice?

What is the plan if treatment is held or stopped?

> **Try explaining the plan back:** Which medicines are you having now? What changes after induction? What would make you call today? Ask your team to fill any gaps.

SOURCE TRANSPARENCY

Where this information comes from

Regimen schedule: NCCP 00455 v16. Medicine explanations and safety information: sources below. The timeline is an educational illustration, not a measured risk curve or an individual prediction.

- NCCP 00455 v16 official regimen
- EMA Keytruda IV product information and patient leaflet
- KEYNOTE-177 randomised first-line colorectal trial
- KEYNOTE-177 final analysis
- Pembrolizumab 400 mg six-week exposure bridging
- HSE treatment safety and urgent advice

Written source addresses

https://healthservice.hse.ie/documents/6903/Pembrolizumab_200mg_Monotherapy_455.pdf

https://www.ema.europa.eu/en/documents/product-information/keytruda-epar-product-information_en.pdf

https://pubmed.ncbi.nlm.nih.gov/33264544/

https://pubmed.ncbi.nlm.nih.gov/35427471/

https://pubmed.ncbi.nlm.nih.gov/32305010/

https://www2.hse.ie/conditions/chemotherapy/side-effects/